Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a Causal Certificate-Driven Single-Cell Digital Twin Platform for Real-Time ATMP Quality Control and Process Optimization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (V), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (V), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-process Technologies	Software	Software Solutions
Concept of Proposed Technology	We develop a deployment-ready, modular software platform for cell & gene therapy (ATMP) process optimization, powered by a single-cell digital twin and causal certificates. A single-cell digital twin is a data-linked simulator that predicts how product quality will respond to interventions (e.g., changing media composition, cytokine dosage, harvest timing) by learning from high-dimensional omics data, rather than only correlations in historical batches. Biomanufacturing faces hidden confounding (donor variability, latent cell states, reagent lot shifts). Our approach produces causal certificates—mathematically guaranteed safety margins on intervention effects—so that operators can make decisions with verified risk bounds even when the true effect is not point-identifiable. The system integrates (i) single-cell transcriptomics or proteomics, (ii) batch process logs, and (iii) periodic QC assays into a unified causal model. Outputs include early-warning risk scores, root-cause pathway explanations, and ranked "next-best" intervention options with audit-ready uncertainty reports, delivering trustworthy AI for regulated environments and reducing false leads in expensive process development.			

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Rationale for Joint R&D Project	ATMP manufacturing is high-value but fragile: small shifts in donor quality, reagent lots, or latent cell states can cause batch failures and costly rework. Korean bio SMEs increasingly generate rich single-cell omics data (transcriptomics, proteomics) during process development, yet struggle to convert this into actionable decisions. Current ML tools are correlation-based and cannot safely guide interventions under hidden confounding (e.g., donor variability confounds media effects). What is missing is a trustworthy, certificate-driven analytics engine that can answer intervention questions (what happens if we change media formulation or harvest timing?) with mathematically verified risk bounds, even when true causal effects are uncertain. To fill this gap, joint R&D is essential: reliable guidance requires (a) real single-cell process data and pilot experiments from the SME to calibrate and validate certificates, and (b) advanced causal inference methods from UIUC to build the digital twin and certificate engine. Phase 1 delivers retrospective analysis and certificate calibration on historical batches. Phase 2 deploys a validated software pipeline into the SME workflow for prospective process optimization, ranking candidate interventions, and reducing false leads in expensive wet-lab validation.
Keywords	1. Single-Cell Digital Twin 2. Causal Certificates 3. ATMP Manufacturing 4. Hidden Confounding 5. Process Optimization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	This project advances ATMP process analytics from detecting quality risks to planning safe, corrective interventions and supporting decision-making. The goal is a deployable software platform that performs near-real-time monitoring, explains why a yield excursion happened, and recommends the "next-best" action—for example, adjusting media composition or harvest timing versus running a specific assay. The recommendations must come with causal certificates so operators can judge whether an action is safe enough to execute despite biological variability. Our group has already built an end-to-end causal inference pipeline that starts from observational data, infers the most plausible causal structure supported by the data, and then estimates causal effects in a sample-efficient way. This is the foundation behind our ICML work [1], where causal effects can be estimated even when the causal structure is only partially known. In this project, we will turn this capability into a causal foundation model for cell manufacturing. The model will ingest single-cell omics data (transcriptomics, proteomics) and streaming bioreactor logs to continuously update a causal graph, which encodes the model's understanding of the physical world (cell metabolism and signaling) and how it is affected by controllable settings (media, DO, pH). This foundation model will act as a "decision-grade process map" that remains valid under donor-to-donor drift and enables rapid root-cause localization and safe action screening during process deviations. For root-cause explanations, we have developed a method for interpreting black-box systems [2] by quantifying how much each input control causally contributes to a quality outcome. This matters in ATMP manufacturing because correlations in historical batches often reflect operator reactions and latent cell states rather than true causal drivers. When cell viability drops due to unmeasured donor variability, operators often adjust feed rates. The logs then show altered feed rates co-occurring with low viability, even though the feed adjustment was a response to the underlying problem and might actually improve viability if donor quality were fixed. A correlation-based tool can therefore recommend incorrect actions during excursions. We will extend causal contribution analysis to cell therapy to identify the most significant causal driver of yield loss (e.g., specific metabolic pathway bottleneck vs. reagent quality). Coupled with the causal foundation model, the system will then simulate candidate interventions on media and process parameters, rank feasible options, and recommend the safest high-impact corrective action under uncertainty. Finally, we will build a causal simulator that remains reliable under hidden confounding, which is prevalent in biomanufacturing because latent cell states, donor variability, and unrecorded reagent shifts affect both process settings and outcomes. We will combine two existing foundations from our lab. The first is scalable sensor-mediated causal effect estimation [3] that stays stable with high-dimensional omics signals. The second is uncertainty-aware causal modeling that produces certified bounds (causal certificates) [4] when latent factors prevent precise point estimates. Together, they enable an interventional generative model for ATMP that simulates quality outcomes under candidate interventions, providing mathematically verified risk bounds for operator planning. In short, the causal foundation model, causal contribution root-cause analysis, and uncertainty-aware intervention simulator together form the certificate-driven single-cell digital twin engine that supports near-real-time QC and beyond-QC decision-making.			

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R&D Target Specifications	**Data readiness and integration**: We will build an end-to-end pipeline that aligns, per batch, (i) process recipes and setpoints, (ii) inline sensor measurements (pH, DO, temp), (iii) downstream QC outcomes (viability, potency), and (iv) single-cell omics profiles. The target is successful alignment for at least 80% of batches in the chosen pilot scope, with missingness patterns and confounding variables explicitly documented in a "data card". **Operational latency**: In shadow-mode operation, the system will output monitoring signals and root-cause explanations at checkpoint cadence (e.g., daily or post-unit operation), meaning updated results are available within one checkpoint cycle. For interactive what-if queries used during process development, the target is response within ≤ 10 seconds for typical queries on the pilot dataset and compute environment. **Monitoring usefulness**: Relative to the SME baseline (current deviation limits or black-box ML), the system will reduce nuisance alarms at a matched detection level on a time-based holdout period. If the baseline lacks a clear metric, we will hold alarm volume fixed and increase the fraction of alarms corresponding to QC-confirmed yield loss. We will report performance with time-split evaluation and uncertainty bounds rather than single-point metrics. **Root-cause explanation quality**: For excursions where a valid cause is known (e.g., maintenance logs, reagent lot change), the system will rank the validated driver within its top-3 suspected drivers in at least 70% of held-out cases. Every explanation will include a causal evidence trail (e.g., specific metabolic pathway or donor factor) and be logged for engineering review. **Action recommendations and feasibility**: For each deviation, the system will output a short list of recommended interventions (e.g., media adjustment, harvest timing). The top-ranked recommendation will be actionable and compliant with SME-provided operating constraints (regulatory ranges and safety rules) in at least 70% of validation cases. Any recommendation violating a hard safety constraint (e.g., maximum cell density) will be automatically blocked. **Uncertainty and trust**: For key decision outcomes (e.g., potency, final yield), the system will provide causal certificates (verified risk bounds). The bounds will be calibrated so that, on held-out data, the realized outcomes fall within the reported bounds at least 90% of the time (nominal coverage), while keeping the bounds practically informative rather than trivially wide. **Pilot impact**: In Phase 2, we will target at least one verifiable improvement in the pilot scope, such as a 15% reduction in batch failure rate or a 20% reduction in time-to-root-cause for deviations. The specific KPI will be selected jointly based on what the SME tracks reliably and can validate during the pilot. **Auditability**: 100% of monitoring outputs, root-cause explanations, and certificate-backed recommendations will be logged with timestamps, input summaries, and model version identifiers to support QA review, traceability, and continuous improvement.
R&D Plan	We propose a 36-month plan: Phase 1 (Months 1–6) foundation and offline calibration, and Phase 2 (Months 7–36) single-cell twin development and pilot deployment. In Phase 1 (Months 0–6), we will achieve data readiness and offline causal prototyping. **Task 1. Data access & problem scoping.** Define the pilot process (e.g., CAR-T expansion) and target QC endpoints. Secure transfer of historical process logs and available single-cell datasets. Establish data governance. **Task 2. Dataset construction & "Data Card".** Build a harmonized retrospective dataset aligning batch recipes, time-series logs, and QC outcomes. Perform missingness analysis and publish a "Data Card"

	explicitly documenting confounding factors (e.g., donor variability). **Task 3. Baselines & evaluation protocol.** Implement SME baselines (univariate SPC, PLS). Define "validated cause" logic from historical deviation reports to serve as ground truth for root-cause ranking. **Task 4. Prototype Causal Foundation Model (Offline).** Apply our ICML structural learning pipeline [1] to historical data to infer the initial causal graph. Separate manipulatable factors (media, feed) from latent drivers. Generate initial "causal certificates" for past interventions to assess coverage. **Task 5. Retrospective Root-Cause Analysis.** Apply causal contribution analysis [2] to historical yield excursions. Compare automated rankings against engineer investigation logs to distinguish true drivers from operator reactions. In Phase 2 (Months 7–36), we will aim to develop a causal digital twin model for single-cell dataset and achieve pilot deployment. **Task 6. Single-Cell Foundation Model Integration.** Ingest high-dimensional single-cell omics to resolve hidden confounding using sensor-mediated estimation [3]. Update the causal graph to act as a "decision-grade process map" that accounts for donor-to-donor variability. **Task 7. Real-Time Certificate Engine.** Implement the uncertainty-aware causal model [4] to generate certified risk bounds at near-real-time cadence. Develop the optimization layer to rank "next-best" actions while filtering out unsafe recommendations. **Task 8. Pilot validation & KPI measurement.** Transition to active pilot advice. Measure reduction in batch failure rate and time-to-root-cause against the baseline period. Validate that 90% of outcomes fall within certificate bounds.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Causal Data Science Lab at UIUC will lead the core algorithmic research and software development for the certificate-driven single-cell digital twin engine, leveraging the lab's expertise in pragmatic causal inference (addressing unmeasured confounding via proxy variables), scalable causal methods for high-dimensional single-cell data, and uncertainty-aware decision-making for safe action recommendations. The lab will design and implement the causal foundation model that estimates intervention effects from observational bioreactor logs while separating true biological drivers from operator responses and latent donor variability. Deliverables include prototype software modules covering the inference engine, explanation layer (causal-contribution root-cause analysis), and operator dashboard with API interfaces. The lab will provide technical documentation, validation reports benchmarking causal ranking against SME baselines, and training sessions to ensure partner engineers can interpret causal certificates and risk bounds.
Expected R&R of Korean SME Partner	The Korean SME partner will provide access to production-grade ATMP manufacturing facilities and the associated data infrastructure, including synchronized datasets of process recipes (media, feed strategies), multi-sensor measurements (pH, DO, temperature profiles), single-cell omics data (transcriptomics/proteomics from development batches), and downstream QC outcomes (viability, potency, sterility). The SME will lead defect taxonomy definition, operational constraint specification (regulatory ranges, safety rules, equipment limits), and pilot testbed governance, including approvals for phased rollout from shadow mode to operator-in-the-loop trials. The SME will implement local data integration, ensuring that UIUC-developed modules can receive near-real-time data streams and push recommendations back into existing visualization systems. The SME will lead operator-in-the-loop validation by coordinating with process engineers and shift supervisors to test recommendations in controlled settings, documenting feasibility and outcomes. Beyond the technical pilot, the SME will take ownership of deployment engineering and the commercialization strategy for broader adoption within the cell therapy CDMO market.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes**: Phase 1 delivers a retrospective single-cell causal foundation model dashboard that supports root-cause diagnosis, what-if querying, and action screening on historical excursions, with validated top-3 root-cause recall of at least 70% and actionable recommendations in at least 70% of cases. Phase 2 delivers a deployed, streaming causal digital twin that performs real-time monitoring, provides uncertainty-calibrated root-cause explanations and intervention recommendations with 90% coverage bounds, and supports operator-in-the-loop trials with full audit logging. **Manufacturing impact**: The system will reduce nuisance alarms while maintaining or improving detection coverage, provide faster root-cause diagnosis with causal evidence trails, and enable safer corrective actions backed by verified risk bounds. In Phase 2 pilot trials, the target is at least one verifiable improvement such as a 15% reduction in batch failure rate, a 20% reduction in time-to-root-cause, or a measurable yield gain on pilot lots, selected jointly with the SME based on what can be reliably tracked. **Commercial and deployment value**: The deliverable is a deployment-ready software module with API and dashboard integration hooks, designed for Korean ATMP/CDMO factories and extensible to related viral vector production, cell line development, and other precision biomanufacturing applications where causal decision support under uncertainty is critical.

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	Broader economic value: Maximizing the yield of high-value cell therapies directly supports national competitiveness in the "Advanced Bio" strategic field. By reducing failure rates in batches that can cost >$500k each, the technology lowers the cost of goods for life-saving treatments, making them more accessible while positioning Korean SMEs as premium, high-tech suppliers in the global biopharma supply chain.

6. References

[1] **Jung, Y.**, Tian, J., and Bareinboim, E. "Estimating identifiable causal effects on Markov equivalence class through double machine learning." International Conference on Machine Learning (ICML), 2021.

[2] **Jung, Y**., Kasiviswanathan, S., Tian, J., Janzing, D., Blöbaum, P., and Bareinboim, E. "On measuring causal contributions via do-interventions." International Conference on Machine Learning (ICML), pp. 10476–10501, 2022.

[3] **Jung, Y.** "Debiased front-door learners for heterogeneous effects." International Conference on Learning Representations (ICLR), 2026.

[4] **Jung, Y.** and Kang, B. "Information-Theoretic Causal Bounds under Unmeasured Confounding." arXiv preprint arXiv:2601.17160v2, 2026.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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